Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Azure Event Hub Connectors 6.4.0

Azure Event Hub connectors allow Stellar Cyber to ingest Azure events and add them to the data lake.

Stellar Cyber connectors with the Collect function (collectors) may skip collecting some data when the ingestion volume is large, which potentially can lead to data loss. This can happen when the processing capacity of the collector is exceeded.

Connector Overview: Azure Event Hub

Capabilities

- Collect: **Yes**
- Respond: **No**
- Native Alerts Mapped: **Yes**
- Runs on: **DP**
- Interval: **Continuous**

Collected Data

Content Type	Index	Locating Records
Azure Activity Log	Syslog	msg_class:
Azure Bastion	Traffic (for Azure Firewall and Structured Azure Firewall Log)	`azure_activity_log`
Azure Firewall	Windows Events (for Microsoft Defender for Cloud)	`azure_bastion_log`
Azure Keyvault		`azure_firewall` (for Azure Firewall and Structured Azure Firewall Log)
Azure Security Center		`azure_keyvault_log`
Azure Security Groups		`azure_security_center`
Azure SQL Server		`azure_security_groups`
Azure Storage		`azure_sql_server`
Azure Synapse Workspace		
Azure Web Application Firewall		
Microsoft Defender for Cloud		

Content Type	Index	Locating Records

Microsoft Graph Activity 6.3.0
Microsoft Sentinel
Structured Azure Firewall Log 6.4.0
		`azure_storage`
		`azure_synapse_workspace`
		`azure_web_application_firewall`
		`microsoft_defender_cloud`
		`microsoft_graph_activity_log` 6.3.0
		`microsoft_sentinel`
		
		msg_origin.source:
		`azure_eventhub`
		`microsoft_azure` (for azure_activity_log)
		
		msg_origin.vendor:
		`microsoft`
		
		msg_origin.category:
		`saas`

Domain

N/A

Response Actions

N/A

Third Party Native Alert Integration Details

Microsoft Defender for Cloud

This connector ingests logs from the content type, Microsoft Defender for Cloud, to get the raw alerts that are stored in the Windows Events index.

Stellar Cyber maps Microsoft Defender for Cloud alerts. The alerts are read from the Windows Events index, enriched with Stellar Cyber fields, and mapped (with deduplication) to the Alerts index.

Deduplication is by `tenantid` and `microsoft_defender_cloud.SystemAlertId`.

For details, see Integration of Third Party Native Alerts.

Microsoft Sentinel

This connector also ingests logs from the content type, Microsoft Sentinel, to get the raw alerts that are stored in the Syslog index.



The integration focuses exclusively on Microsoft Sentinel Incident data and only supports data transferred through Azure Event Hub.

Stellar Cyber maps Microsoft Sentinel incidents. The alerts are read from the Syslog index, enriched with Stellar Cyber fields, and mapped (with deduplication) to the Alerts index.

Deduplication is by `tenantid` and `microsoft_sentinel.IncidentName`.

For details, see Integration of Third Party Native Alerts.

Required Credentials and Configurations

- Event Hub Name, Connection String, Consumer Group

Locating Records

Microsoft Defender for Cloud

To search the alerts in the Alerts index or to search the Original Records in the Syslog index, use the query: `msg_class: (microsoft_defender_cloud OR AzureSecurityCenter)`.

Microsoft Sentinel

To search the alerts in the Alerts index or to search the Original Records in the Syslog index, use the query: `msg_class: microsoft_sentinel AND microsoft_sentinel.Type: SecurityIncident`.

Adding an Azure Event Hub Connector

To add an Azure Event Hub connector:

1. Gather the Event Hub information
2. Stream logs to Event Hub
3. Add Azure Monitor logs to Event Hub
4. Add the connector in Stellar Cyber
5. Test the connector
6. Verify ingestion

Gathering the Event Hub Information

Gather the following Event Hub information.

The subscription plan needed for the Azure Event Hub connector is Basic.

- **Azure Event Hub Name**: The name of the Azure Event Hub.
- **Connection String**: Find the connection string in Azure. You must use a unique connection string for each of your Stellar Cyber Azure Event Hub connector instances.

  The connection string must include the **Listen** permission for shared access policies.

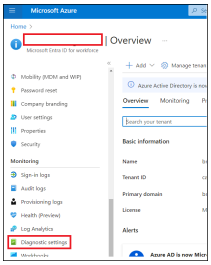
- **Consumer Group**: The consumer group for the Event Hub.
- **Event Source**: The source of the events you want to ingest (you must configure these sources to send data to your Event Hub):
  - Azure Activity Log
  - Azure Bastion
  - Azure Firewall
  - Azure Keyvault
  - Azure Security Center

- Azure Security Groups
- Azure SQL S r (includes AuditEvent log)
- Azure Storage
- Azure Synapse Workspace
- Azure Web Application Firewall
- Microsoft Defender for Cloud
- Microsoft Graph Activity Log 6.3.0
- Microsoft Sentinel
- Structured Azure Firewall Log 6.4.0
- Custom Event Source

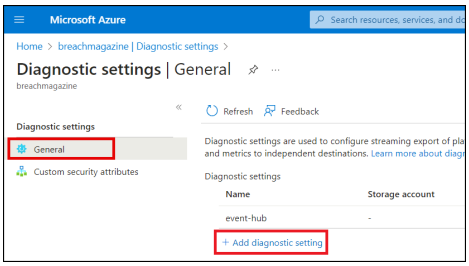
Streaming Logs to Event Hub

To stream logs to an Event Hub, refer to the Microsoft documentation.

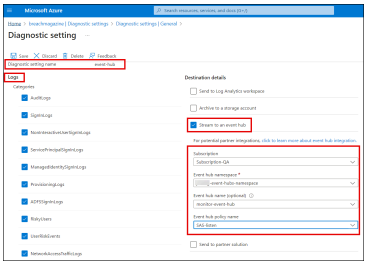
1. Log in as a Security Administrator to Microsoft Entra ID.



2. Scroll down to **Monitoring** and click **Diagnostic settings**.
3. Click **General** and then click **Add diagnostic setting**.



4. In the **Diagnostic setting** window:



   a. Enter a **Diagnostic setting name**.
   b. Select the **Logs** you want to stream.
   c. Select the check box for **Stream to an event hub**.

d. For where you want to route the logs, select the **Subscription**, **Event hub namespace**, optional **Event hub name**, and **Event hub policy name**.

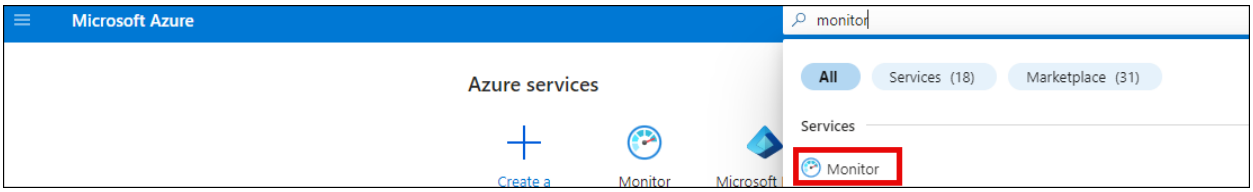
5. Click **Save**.

Make sure the **Subscription** and **Event hub namespace** are associated with the Microsoft Entra tenant from where the logs are streamed.

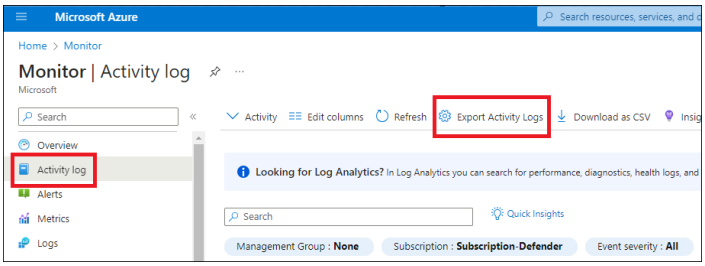
Adding Azure Monitor Logs to Event Hub

To add Azure monitor logs to an Event Hub:

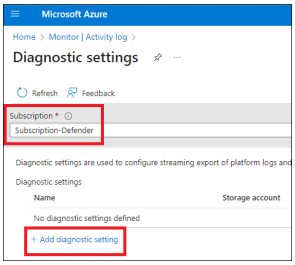
1. In Microsoft Azure, portal.azure.com, search for and select the Monitor service.



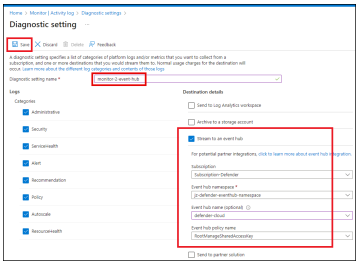
2. Click **Activity log** and then click **Export Activity Logs**.



3. For the **Diagnostic settings**, select a **Subscription** and then click **Add diagnostic setting**.



4. In the **Diagnostic setting** window:



a. Enter a **Diagnostic setting name**.

b. Select the **Logs** you want to stream.

c. Select the check box for **Stream to an event hub**.

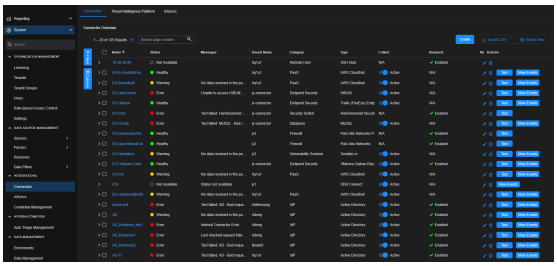
d. For where you want to route the logs, select the **Subscription**, **Event hub namespace**, optional **Event hub name**, and **Event hub policy name**.

5. Click **Save**.

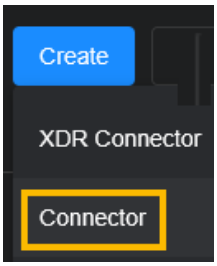
Adding the Connector in Stellar Cyber

With the access information handy, you can add an Azure Event Hub connector in Stellar Cyber:

1. Log in to Stellar Cyber.
2. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.

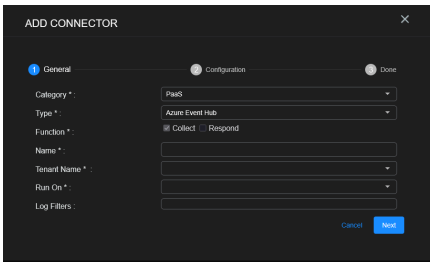


3. Click **Create** and select **Connector**.



The **General** tab of the **Add Connector** screen appears. The information on this tab cannot be changed after you add the connector.

4. Choose **PaaS** from the **Category** dropdown.
5. Choose **Azure Event Hub** from the **Type** dropdown.



The asterisk (*) indicates a required field.

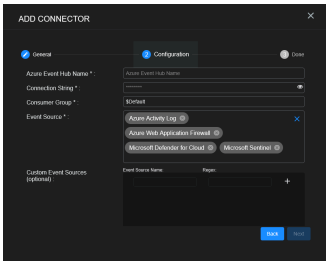
6. For this connector, the supported **Function** is **Collect**, which is enabled already.
7. Enter a **Name**.

Notes:

- This field does not accept multibyte characters.
- It is recommended that you follow a naming convention such as tenantname-connectortype.

8. Choose a **Tenant Name**. The Interflow records created by this connector include this tenant name.
9. Choose the device on which to run the connector.

10. (Optional) When the **Function** is **Collect**, you can apply **Log Filters**. For information, see Managing Log Filt
11. Click **Next**. The **Configuration** tab appears.

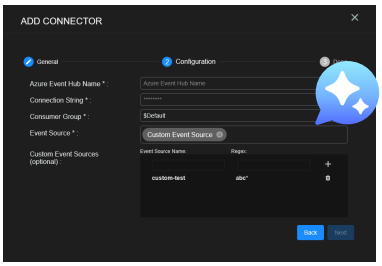


The asterisk (*) indicates a required field.

12. Enter the **Azure Event Hub Name** you noted above.
13. Enter the **Connection String** you noted above.
14. Enter the **Consumer Group** you noted above.
15. Select one or more **Event Source**. The available sources are:
    - Azure Activity Log
    - Azure Bastion
    - Azure Firewall
    - Azure Keyvault
    - Azure Security Center
    - Azure Security Groups
    - Azure SQL Server (includes AuditEvent log)
    - Azure Storage
    - Azure Synapse Workspace
    - Azure Web Application Firewall
    - Microsoft Defender for Cloud
    - Microsoft Graph Activity Log 6.3.0
    - Microsoft Sentinel
    - Structured Azure Firewall Log 6.4.0

      The order of the event sources in the above list dictates the content type that will be matched first. The event source that was added first will match first.

    - Custom Event Source (choose only if using optional **Custom Event Sources**)
16. (Optional) Use **Custom Event Sources** to set up customized event sources that pull specific types of data from Azure Event Hub.



For guidelines on how to create a custom content type, see Creating a Custom Content Type.

a. Select **Custom Event Source** from the **Event Source** dropdown menu.

b. In the **Custom Event Sources (optional)** area, enter a name for the **Event Source Name** that is unique within the connector. The name will be used for setting the `msg_class` in the following format: `azure_<customer defined name>`.

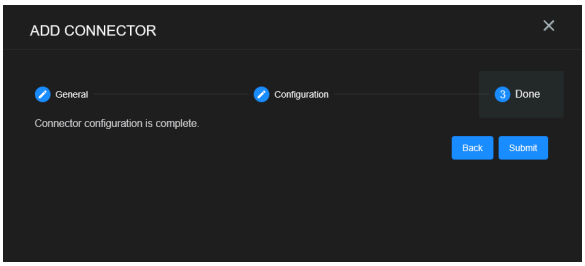
c. Enter a **Regex**, a Regular Expression that defines a unique string expected in the data, for example, `abc*`. The connector looks in Azure Event Hub and pulls any data that matches the Regex.

d. Click the + sign to add another custom event source. The recommended maximum is 10.

The order of the event sources in the list dictates the content type that will be matched first. The custom event source that appears in the list at the top will match first, meaning that the one added last will match first.

e. Click the trash can to delete a custom event source.

17. Click **Next**. The final confirmation tab appears.



18. Click **Submit**.

The new connector is immediately active.

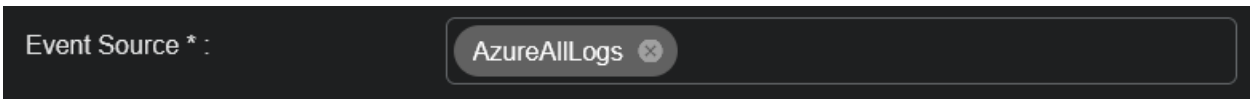
Creating a Custom Content Type

The following are guidelines to create a custom content type using **Custom Event Sources** in the Azure Event Hub connector.

- Collect the log sample from Azure Event Hub directly to understand the log content, because the Stellar Cyber data enrichment and normalization occur after the custom parser matching of the configured Regex from the raw data retrieved from Azure Event Hub.

- If you are not able to do the above step, then you can use the **AzureAllLogs** content type to study the data content.

  For debugging purposes, you can optionally type **AzureAllLogs** into the **Event Source** dropdown menu and press Enter. The content type will display as follows:

  See the following AzureAllLogs Format Limitation.

  The **AzureAllLogs** content type collects all logs available to the configured credentials. The logs collected this way have the `msg_class: azure_all`.

  Keep in mind that **AzureAllLogs** is a debugging content type and is not usually used by the customer in normal cases. It may cause conflicting field types in Elasticsearch and thus data loss, because all logs that do not match the existing parser will send data under this content type.

- To create the custom content type:

  a. Select **Custom Event Source** from the **Event Source** dropdown menu.

  b. In the **Custom Event Sources (optional)** area, enter a name for the **Event Source Name** that is unique within the connector.

  c. Enter a unique **Regex** applied to this custom event source only. It can be either a field or value because each event is a string (`body_as_str`) before it is converted to a valid JSON object. So use a unique value, a unique field, or both, or multiple values/fields to identify the event source. For information on `body_as_str`, see the Microsoft documentation.

- To verify the logs ingested under the custom event source, use the `msg_class` as an indicator. The `msg_class` for a custom event source has the following format: `azure_<customer defined name>`, for example, `azure_<source_name>`.

AzureAllLogs Format Limitation

The data from Azure sources is expected to have the following format:

```
{'records': [...]}
```

which is an object with the key 'records' and the value as a list of objects.

The schema for data coming from another source with another format cannot be guaranteed to be parsed correctly.

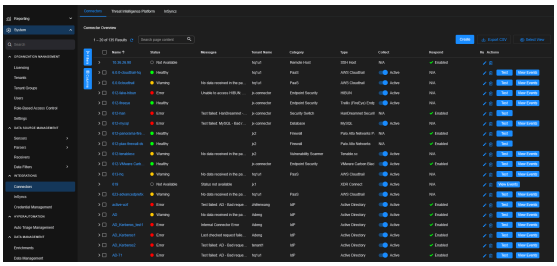
Check your source of Azure Event Hub data and use the expected format in the raw event logs.

Testing the Connector

When you add (or edit) a connector, we recommend that you run a test to validate the connectivity parameters you entered. The test validates authentication and connectivity.

For connectors running on a sensor, Stellar Cyber recommends that you allow 30-60 seconds for new or modified configuration details to be propagated to the sensor before performing a test.

1. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.



2. Locate the connector by name that you added, or modified, or that you want to test.
3. Click **Test** at the right side of that row. The test runs immediately.

Note that you may run only one test at a time.

Stellar Cyber conducts a basic connectivity test for the connector and reports a success or failure result. A successful test indicates that you entered all of the connector information correctly.

To aid troubleshooting your connector, the dialog remains open until you explicitly close it by using the **X** button. If the test fails, you can select the button from the same row to review and correct issues.

The connector status is updated every five (5) minutes. A successful test clears the connector status, but if issues persist, the status reverts to failed after a minute.

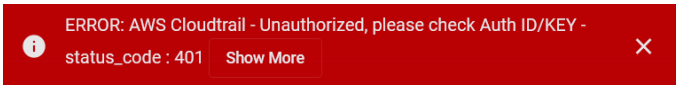
Repeat the test as needed.

Display sample messages...

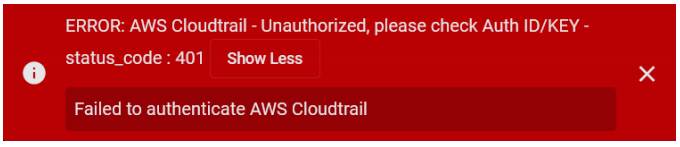
Success !



Failure with summary of issue:



Show More example detail:



If the test fails, the common HTTP status error codes are as follows:

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
400	Bad Request	This error occurs when there is an error in the connector configuration.	Did you configure the connector correctly?

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
401	Unauthorized	This error occurs when an authentication credential is invalid or when a user does not have sufficient privileges to access a specific API.	Did you enter your credentials correctly? Are your credentials expired? Are your credentials entitled or licensed for that specific resource?
403	Forbidden	This error occurs when the permission or scope is not correct in a valid credential.	Did you enter your credentials correctly? Do you have the required role or permissions for that credential?
404	Not Found	This error occurs when a URL path does not resolve to an entity.	Did you enter your API URL correctly?
429	Too Many Requests	This error occurs when the API server receives too much traffic or if a user's license or entitlement quota is exceeded.	The server or user license/quota will eventually recover. The connector will periodically retry the query. If this occurs unexpectedly or too often, work with your API provider to investigate the server limits, user licensing, or quotas.

For a full list of codes, refer to HTTP response status codes.

Verifying Ingestion

To verify ingestion:

1. Select **Threat Hunting**.

   The **Interflow Search** tab appears.

2. Change the **Indices** for the type of content you collected:

   - For the majority of content types, change the **Indices** to **Syslog**.
   - For **Microsoft Defender for Cloud** only, change the **Indices** to **Windows Events**.
   - For **Azure Firewall** and **Structured Azure Firewall Log** only, change the **Indices** to **Traffic**.

   The table immediately updates to show ingested Interflow records.

For custom event sources, the unique name provided during configuration displays in **Threat Hunting** with a `msg_class` containing an `azure` prefix. For example, if the unique name is `test`, the `msg_class` in **Threat Hunting** is `azure_test`.



© 2026 Stellar Cyber . All rights reserved.